Please attach this cover sheet to your completed homework

Homework 2 (Ch. 8) **ECE 311 – Hang Gao**

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Problem	Course outcome	Grade
8.1	1.c.	/36
8.2	1.c.	/20
8.5	1.c.	/15
8.6	1.c.	/15
8.11	1.c.	/14
	TOTAL:	/100

ABET Assessed Outcomes

1.c. Use appropriate models to formulate solutions related to electric circuit problems.

Homework #2
Due date: Sunday, February 8th by 11:59 PM (Electronically – Canvas)

8.1 A Y-connected balanced three-phase source is feeding a balanced three-phase load. The voltage and current of the source coil are

$$v(t) = 340\sin(377t + 0.5236)V$$

$$i(t) = 100\sin(377t + 0.87266)A$$

The phase shift angles in the two equations are in radians. Calculate the following:

a. The rms phase voltage
b. The rms line-to-line voltage
c. The rms current in the source
d. The rms current in the transmission line
e. The frequency of the supply
f. The power factor at the source side, state leading or lagging
g. The three-phase real power delivered to the load
h. The three-phase reactive power delivered to the load
i. If the load is connected in delta configuration, calculate the load impedance

8.2 The current and voltage of a wye-connected load are:

$$\overline{V}_{ca} = 480\angle -60°V$$

$$\overline{I}_b = 20\angle 120°A$$

a. Compute $\overline{V}_{an}$
b. Compute $\overline{I}_a$
c. Compute the power factor angle
d. Compute the real power of the load

8.5 A balanced wye-connected load of (4+j3)Ω is connected across a three-phase source of 173 V (line-to-line).

a. Find the transmission line current
b. Find the power factor, the complex power, the real power, and the reactive power of the load.
c. With V_{ab} as the reference, sketch the phasor diagram that shows all voltages and currents.

8.6 Two three-phase wye-connected loads are in parallel across a three-phase supply. The first load draws a current of 20 A at 0.9 power factor leading, and the second load draws a current of 30 A at 0.8 power factor lagging. Calculate the following:

a. The transmission line current

b. The power factor of the load

c. The real power supplied by the source if the line-to-line voltage of the transmission line is 400 V

8.11 A three-phase wye-connected source is energizing a delta-connected load. The phase voltage of the transmission line is $\overline{V}_{bn} = 120\angle 0°V$, and the total load impedance is $\overline{Z} = 9\angle 30°\Omega$. Compute the line current $\overline{I}_a$ and the real power consumed by the delta load.